

PHU DUONG

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EDUCATION

Princeton University

*Bachelor of Science in Engineering in Computer Science
Minor in Statistics and Machine Learning*

Expected May 2028

Princeton, NJ

- Relevant Coursework: Algorithms and Data Structures, Advanced Algorithms, Machine Learning, Natural Language Processing, Systems Programming, Computer Architecture, Operating Systems, Statistics, Probability and Stochastic Systems
- Awards: QuestBridge National College Match Scholarship

EXPERIENCE

Princeton Computer Science

Teaching Assistant

September 2025 – Present

Princeton, NJ

- Tools: Java, C
- Led office hours for **175+ students** in COS 226: Algorithms and Data Structures, teaching searching, sorting, and graph algorithms, debugging Java implementations, and analyzing runtime complexity
- Led office hours for **120+ students** in COS 217: Introduction to Programming Systems, teaching C, ARMv8 assembly, and Linux systems, debugging segmentation faults and memory leaks with GDB and Valgrind

NashTech

Software Engineer Intern

June 2025 – August 2025

Ho Chi Minh City, VN

- Tools: PHP, JavaScript, PostgreSQL, AWS
- Developed an assignment management system for The Open University, aggregating coursework data across **600+ courses** for **220,000+ students and educators**, centralizing tracking and administration previously scattered across individual modules
- Optimized assignment data queries to support **60,000+ daily active users** by identifying bottlenecks with Amazon CloudWatch, implementing composite indexes, and rewriting SQL queries to reduce query execution time by **~50%**
- Implemented **100+** PHPUnit unit and Behat integration tests in a CI/CD pipeline, automating pull request validation and preventing regressions from reaching production

Jane Street

Academy of Math and Programming Scholar

June 2024 – August 2024

New York, NY

- Tools: Python
- Developed a high-frequency trading algorithm implementing statistical arbitrage and mean reversion strategies, generating **\$5,000+ PnL** per round in Jane Street's Electronic Trading Competition
- Built optimization algorithms for strategy games, using Monte Carlo simulations to calculate expected values for Camel Up bets and a minimax strategy to solve Wordle in **~3.7 guesses**
- Ranked **1st place** among teams totaling 78 peers in Jane Street's Estimathon and Mystery Planet math and trading competitions

MIT Momentum AI

Capstone Scholar

August 2022 - August 2022

Cambridge, MA

- Tools: PyTorch
- Developed a convolutional neural network to classify reCAPTCHA images, training it on a **12,000-image** Google reCAPTCHA V2 dataset, and achieving **~90% accuracy**

PROJECTS

Natural Language Circuit Component Selector | TypeScript, Python, React, Flask

- Developed an LLM-based tool that automates PCB component selection and validation, reducing design time from days to minutes by analyzing natural language requirements
- Orchestrated agents to build a component connection graph using datasheet specifications, selecting the most restrictive components first and validating electrical/physical compatibility against existing selections
- Built a Model Context Protocol server with tools for component search and specification retrieval, enabling agents to query a component database of **10,000+ parts**
- **Won Best Business and Enterprise Hack** among 194 projects and 600+ participants at HackPrinceton

Biometric-Controlled Melatonin Patch | TypeScript, C++, React Native, Node.js, Express

- Developed a web/mobile app to optimize sleep cycles by collecting biometric data from the Fitbit API to control automated melatonin delivery through a connected transdermal patch
- Built a linear regression model to personalize melatonin dosing during sleep based on heart rate, respiratory rate, and user sentiment extracted via the Google Gemini API
- Engineered a Wi-Fi-enabled dispenser using Arduino and ESP8266, programming embedded C++ to process server requests and control melatonin release
- Achieved **2nd place** out of 87 projects in the Hardware category at HackPrinceton

SKILLS

Languages: Java, Python, C/C++, JavaScript/TypeScript, PHP, HTML/CSS, SQL

Frameworks & Technologies: PyTorch, Node.js, Express, Flask, React, React Native, Docker, Linux, Git